

JEE MAIN REVISION MASTER: FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 36 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Domain, Range & Composite Functions	22 PYQs	Domain & Range
High	Injective (One-One), Surjective (Onto) & Inverse	14 PYQs	Mappings

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Domain, Range & Composite Functions	$\text{For } \sqrt{f(x)} \implies f(x) \geq 0, \quad \text{For } \log_b(f(x)) \implies f(x) > 0, b > 0, b \neq 1, \quad x = [x] + \{x\}$
Injective (One-One), Surjective (Onto) & Inverse	$\text{One-One: } f'(x) > 0 \text{ (strictly monotonic)}, \quad \text{Onto: } \text{Range}(f) = \text{Codomain}(f), \quad \text{Bijective} \iff \text{Invertible}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Domain, Range & Composite Functions (Very High Yield — 22 PYQs)

Core Principle: Determination of real domain and range, floor and fractional part functions, and domain constraints on composite functions.

■ **Caution / JEE Pitfall:** Domain of $f(g(x))$ is NOT simply the domain of g ; it requires $x \in \text{Domain}(g)$ AND $g(x) \in \text{Domain}(f)$!

■ **Speed Trick:** The range of $f(x) = a \sin x + b \cos x + c$ is $[c - \sqrt{a^2 + b^2}, c + \sqrt{a^2 + b^2}]$.

Representative Source PYQs:

- [2025 Shift] Q#3 (MCQ): "Domain of the function f , given by $f(x) = \frac{1}{\sqrt{(x-2)(x-5)}}$ is..." [Ans: N/A]
- [2025 Shift] Q#4 (MCQ): "If $f(x) = \sin[\pi^2]x - \sin[-\pi^2]x$, where $[x]$ = greatest integer $\leq x$, then which of the following is not true?..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If $[x]^2 - 5[x] + 6 = 0$, where $[x]$ denotes the greatest integer function, then..." [Ans: N/A]

3.2 Injective (One-One), Surjective (Onto) & Inverse (High Yield — 14 PYQs)

Core Principle: Horizontal line test, monotonicity test for injectivity, codomain matching for surjectivity, and finding function inverses.

■ **Caution / JEE Pitfall:** A function cannot be invertible unless it is BOTH one-one and onto (bijective).

■ **Speed Trick:** If $f(x)$ is continuous and strictly increasing ($f'(x) > 0$) or strictly decreasing on $\mathbb{R}$, it is always one-one.

Representative Source PYQs:

- **[2026 Shift] Q#1 (MCQ):** " $f(x)=(x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ is..." **[Ans: C]**
- **[2026 Shift] Q#2 (MCQ):** "The domain of the function $\sqrt{(x-7/9-x)}$ is..." **[Ans: B]**
- **[2025 Shift] Q#1 (MCQ):** "Let the functions " f " and " g " be $f:[0,(\pi/2)] \rightarrow \mathbb{R}$ given by $f(x)=\sin x$ and $g:[0,(\pi/2)] \rightarrow \mathbb{R}$ given by..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$f(x)=(x-1)^2$ for $x \geq 1$, $g(x)$ is a function whose graph is the reflection of the graph of $f(x)$ in the line $y=x$, then $g(x)$ is

- (A) $-\sqrt{x}+1$
- (B) $\sqrt{x}-1$
- (C) $\sqrt{x}+1$ **[Correct]**
- (D) $\sqrt{x-1}$

Correct Answer: C

Solution Analysis: The graph of $g(x)$ is the reflection of the graph of $f(x)$ in the line $y=x$. Reflection in the line $y=x$ gives the inverse relation of the function. Given, $f(x)=(x-1)^2, x \geq 1$. Let $y=(x-1)^2$. Since $x \geq 1$, we have $x-1 \geq 0$. So when we take square root, $\sqrt{y}=x-1$. Hence, $x=\sqrt{y}+1$. Now interchange x and y to get the reflected function: $y=\sqrt{x}+1$. So, $g(x)=\sqrt{x}+1$. Hence the correct option is Option C $\sqrt{x}+1$.

Benchmark Problem 2: KCET 2024 (Q#2)

If $[x]^2 - 5[x] + 6 = 0$, where $[x]$ denotes the greatest integer function, then

- (A) $x \in [3, 4]$
- (B) $x \in [2, 4]$
- (C) $x \in [2, 3]$
- (D) $x \in (2, 3]$

Correct Answer: None

Solution Analysis: $[x]^2 - 5[x] + 6 = 0 \Rightarrow [x]^2 - 3[x] - 2[x] + 6 = 0 \Rightarrow [x]([x]-3) - 2([x]-3) = 0 \Rightarrow ([x]-3)([x]-2) = 0 \Rightarrow [x]=2 \text{ or } 3$.
 $[x]=2 \Rightarrow x \in [2, 3)$ or $[x]=3 \Rightarrow x \in [3, 4)$. Hence, $x \in [2, 4)$.